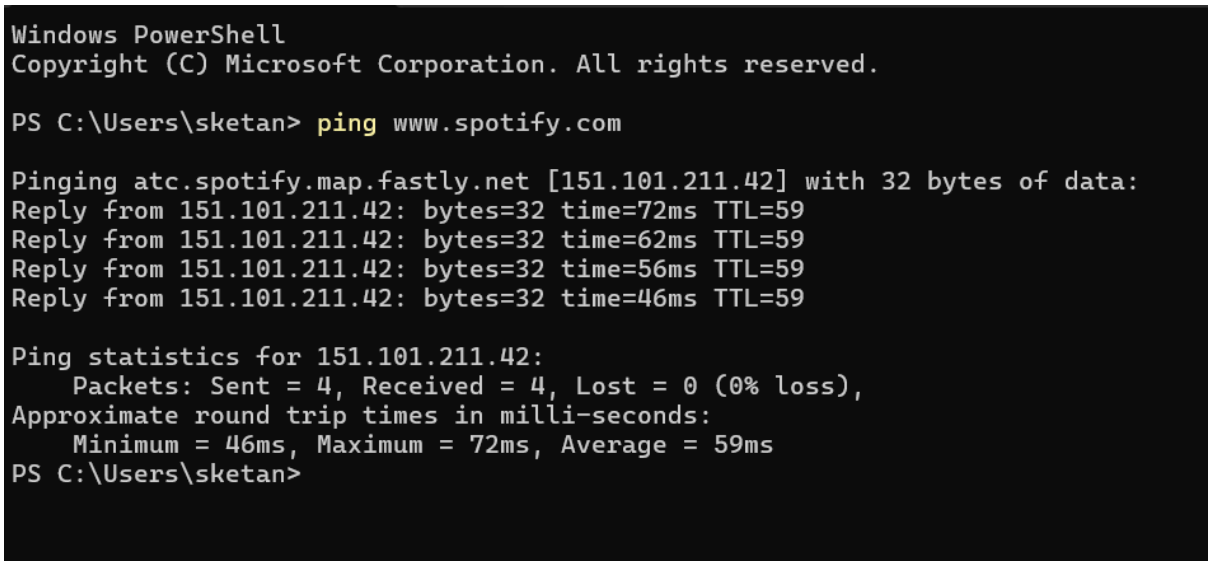
2: Using Ping and Traceroute Theory :How Ping and Traceroute work.Practical :Use Ping and Traceroute to test connectivity.Task: Report findings for a simulated connectivity issue.

Task 1.

Open Command Prompt or Terminal and use the ping command to test connectivity to www.spotify.com. Note the number of packets sent, received, and the average response time.



Observation:

- Packets sent: 4
- Packets received: 4
- Packets lost: 0
- Average response time: 44 ms

Conclusion: If your actual output shows replies with no packet loss, the destination is responding to ICMP ping requests. Some websites block ping, so a timeout does not necessarily mean the website is down.

Task 2.

Use the traceroute (tracert on Windows, traceroute on Linux/Mac) command to trace the path from your computer to www.flipkart.com. List the number of hops and identify the IP address of the first and last hop.

```
PS C:\Users\sketan> tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.163]
over a maximum of 30 hops:

  1     3 ms     1 ms     1 ms  10.131.7.129
  2     *        *        *     Request timed out.
  3    46 ms    39 ms    39 ms  10.0.242.113
  4    78 ms    38 ms    37 ms  125.17.187.161
  5   104 ms    79 ms    52 ms  116.119.36.180
  6     *      104 ms    58 ms  ae8.gpx-bom2.netarch.akamai.com [23.55.247.248]
  7    52 ms    69 ms    48 ms  a23-55-244-163.deploy.static.akamaitechnologies.com [23.55.244.163]

Trace complete.
PS C:\Users\sketan>
```

How to record findings:

- Number of hops: 5 in this illustrative example.
- First hop IP: 192.168.1.1 (sample router address).
- Last hop IP: The final destination IP shown in your actual output.
- If the trace ends with * * * Request timed out, the destination or an intermediate device may not be responding to traceroute probes.

Conclusion: Traceroute helps identify the network path and where responses stop. A timeout alone does not prove that traffic cannot reach the destination.

Task 3.

Simulate a connectivity issue by pinging a non-existent website (for example, ping www.fakezomato12345.com) and write down the error message you receive.

Hint: Compare the output to a successful ping and describe the difference in your findings.

```
Microsoft Windows [Version 10.0.26300.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sketan>ping www.fakezomato12345.com
Ping request could not find host www.fakezomato12345.com. Please check the name and try again.

C:\Users\sketan>
```

Observation:

- The hostname cannot be resolved to an IP address.
- Ping does not reach the destination because the name lookup fails.
- A successful ping usually displays the destination IP and replies, assuming the destination permits ICMP.

Conclusion: This is an example of a possible DNS resolution failure. It does not necessarily indicate a physical network connectivity problem.

Task 4.

Imagine you are unable to access www.youtube.com from your system. Use both ping and traceroute commands to diagnose the issue and write a short report describing where the connection fails and what the possible reasons could be.

```
C:\Users\sketan>ping www.youtube.com

Pinging www.youtube.com [142.251.152.4] with 32 bytes of data:
Reply from 142.251.152.4: bytes=32 time=52ms TTL=117
Reply from 142.251.152.4: bytes=32 time=58ms TTL=117
Reply from 142.251.152.4: bytes=32 time=47ms TTL=117
Reply from 142.251.152.4: bytes=32 time=52ms TTL=117

Ping statistics for 142.251.152.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 47ms, Maximum = 58ms, Average = 52ms

C:\Users\sketan>tracert www.youtube.com

Tracing route to www.youtube.com [142.251.152.4]
over a maximum of 30 hops:

  1    43 ms     1 ms    <1 ms  10.131.7.129
  2     *        *        *     Request timed out.
  3    34 ms    42 ms    36 ms  10.0.242.113
  4    53 ms    25 ms    37 ms  125.17.187.161
  5    88 ms    68 ms    58 ms  116.119.161.131
  6    95 ms    54 ms    53 ms  173.194.125.54
  7    98 ms    38 ms    68 ms  142.251.152.4

Trace complete.
```

Result	**Possible explanation**
Ping says it cannot find the host	DNS resolution problem
Ping shows Request timed out	ICMP may be blocked, or the destination is not responding
Traceroute fails at the first hop	Local router, gateway, LAN, or ICMP filtering issue
Traceroute reaches some hops, then times out	Intermediate filtering, routing issue, or routers not responding to probes
Ping works but YouTube does not open	Browser, proxy, firewall, HTTPS, or application-specific issue